

Thesis writing milestones

Neural Data Science Lab (Prof. Alexander Ecker), eckerlab.org
Computational and Quantitative Neuromuscular Imaging Group (Dr. Janek Gröhl)
Optimal Transport Group (Prof. Bernhard Schmitzer), ot.cs.uni-goettingen.de
Neuronal Intelligence Group (Prof. Fabian Sinz), sinzlab.org

Last update: March 2025

OVERVIEW

The purpose of the three thesis writing milestones and accompanying tutorial sessions is to help you structure your thesis work and provide guidance (where necessary) on how to approach the various phases of thesis work from the initial proposal stage, over carrying out the research to writing it up.

This document describes what we as the people running the tutorials expect you to deliver for the three milestones. Both the person writing the thesis (referred to below as ‘author’) and the writing buddy should carefully read this information and plan accordingly. Both should be present in the tutorial.

We expect each writing buddy to invest on the order of a full day on the other’s thesis for each milestone. Please reserve this time in your schedule early on and plan accordingly. You can expect to receive the same amount of work invested in your own thesis.

The tutorials and milestones complement the supervision and guidance you are getting from your primary supervisor and should in no way replace it or work against it. Please discuss the timeline and instructions with your primary supervisor at the beginning of your thesis work. If there are any questions or disagreements, please touch base early on with us to discuss it.

MILESTONE 1: RESEARCH PROPOSAL

We ask you to complete the following tasks:

1. Follow the research proposal template below to create the first outline of your own thesis proposal. A good research proposal has around 3–4 pages (excluding title and reference list)
 - a. Take a look at the following thesis proposal examples to help you with this assignment: [Example 1](#), [Example 2](#)
2. Before writing the section “problem statement”, know what makes a good research question and how to define it. Watch [this video](#) to check if your research question(s) meet the requirements.

Template: Research Proposal [author]

Student Name: XXX

Supervisor(s), organization unit, institution: XXX

Date of this Document: XX.XX.20XX

Introduction

- Context: Give a brief overview of the problem area.
- State of the Art and related work: Briefly discuss the approaches and results that have already been presented in this or related areas (at least the five most important papers). This can also be a separate section

Goal of the Project/Thesis

Problem Statement

- What is the problem that your project/thesis addresses?
 - The problem should logically follow from the description of the state of the art. Clearly describe the need for the research you plan to perform (research gap).
 - What is unknown? What has been ignored or understudied? What could be improved? what are the limitations of current solutions (if solutions already exist)?
- Why is it a significant (interesting/important) problem?

Research Objective(s)

- Which specific **research question(s)** will you try to answer?
- **OR:** Which particular problem(s) will you try to solve?
 - Which sub-problems do you derive and address in your project/thesis?
- **OR:** Which hypotheses will you investigate?

Expected Results

- Which specific results shall the project achieve? Clearly describe your expected contribution:
 - What will be known afterwards that is not known now? / What will be created that does not exist now? Whom will this benefit? Why is it important?
 - Provide quantitative estimates for your expected results. For instance, you may expect your model to achieve an accuracy of 90% on task X, or variables A and B to have a correlation of 0.7, etc. Discuss with your supervisor what they expect you to find.

Methodology

- Which specific tasks will you perform to achieve the objectives and expected results?
- What methods will you apply to accomplish each task, e.g., literature research, empirical study, prototype implementation, quantitative or qualitative evaluation etc.?
- How will you evaluate the fulfillment of your research objective(s)? How will you show that you answered the question, measure how well you solved the problem, or prove that the hypothesis held? For an example while writing this chapter of your thesis, read Chapter 6 of [this thesis](#) and Chapter 8 of this book.
- Which tools, datasets etc. do you plan to use?

Work plan

- Create a Gantt chart that shows your planned research tasks (defined in the previous section) and their approximate start and end dates.

References

- List relevant sources (state of the art and related work)
- Search for related academic research/ projects using [Google Scholar](#) and [Semantic Scholar](#)
- Computer Science Digital Libraries: [ACM](#), [IEEE](#)

Peer feedback [writing buddy]

Please look at your buddy's research proposal and provide feedback along the following questions. We will also discuss these questions in the meeting:

- Introduction
 - Is the overall motivation clear and does it provide the necessary background?
 - Is the related work discussed?
- Goal of the thesis
 - What is the gap in knowledge / research goal?
 - How does the approach differ from/improve upon related/prior work?
 - Are specific objectives spelled out?
 - What are the expected results and contributions?
- Methodology
 - Is the approach described clearly?
 - Are specific work packages spelled out?
 - Is it clear from the description of the work packages how much work they entail?
 - Is the research plan realistic to accomplish in the given timeframe?

MILESTONE 2: OUTLINE, FIGURES & TABLES DRAFT

Here we create the scaffolding of your thesis document, which you will be filling over the remaining weeks. This is your first (very rough) thesis draft. At this point you should know

- What the messages are you want to convey to your reader
- How to provide evidence to support your statements.

We ask you do complete the four tasks outlined below until the meeting.

Thesis outline [author]

Create an outline of your thesis defining the main sections (e.g. introduction, related work, ...) and their subsections and discuss it with your supervisor. Copy & paste the introduction and related work parts from your research proposal into this document. It forms a first draft you can revise over the next couple of weeks (you do not need to revise it yet). For the main part containing methodology and results, follow the steps below.

Results: subsections and topic sentences [author]

Define the **subsections** and write the **topic sentences** for your results section. A topic sentence is a specific statement that describes a result. For instance, a topic sentence could be

The MYFANCYALGORITHM network model outperforms state of the art CNNs in image classification

A topic sentence does not need any evidence but should specifically state *what* you find. If you don't have results for that part of your thesis yet, state what you expect to find.

Think of the topic sentences as the first sentence of each paragraph in your results section. Later you will provide/describe the evidence to support this statement in the remaining paragraph.

The sequence of all topic sentences should yield a consistent picture of what you found in your thesis. If a reader reads only the topic sentences, they should be able to tell someone else what you did.

Results: draft figures and tables [author]

The topic sentences provide the logical structure of your argument. Now you need to provide evidence for these statements. This evidence will eventually need to be provided by **figures** and **tables**. Think about each topic sentence and list the evidence you need to convince a reader that your topic sentence is "true." Make a bullet point list of the evidence you need behind the topic sentence. For the example above, this could be

The accuracy of MYFANCYALGORITHM needs to be significantly higher than that of control CNN models (bar plot, table, t-test).

Once you have that, draft the figures you need to provide the evidence. If you already have the figures or draft versions, put them in. If you don't have them yet, draw (literally, with a pen) what you expect them to look like.

For each figure, write the figure caption. This should briefly and succinctly describe what is shown in the figure and what the take home message from this is. For example:

Figure 1: Model comparison. A) mean accuracy for MYFANCYALGORITHM and control model. Errorbars denote standard error of the mean (n=2000). The mean accuracy of MYFANCYALGORITHM is significantly better (paired t-test, $t=...$, $p<10^{-3}$). B) ...

When you draft the figures, keep in mind that a reader should be able to get a good idea of your work by looking only at the figures, tables and captions.

Methodology: sections and subsections [author]

Given the results and supporting evidence you show, think about which methods, models, experiments, datasets etc. you used. They will be described in appropriate sections typically placed before the results. As there are different approaches to structuring papers and theses, discuss with your supervisor how to structure this part of your thesis. Define the sections and subsections.

Methodology: figures and tables [author]

Go over the sections and subsections describing your methodology and think about which aspects require or benefit from visualizations. Examples include (but are not limited to) schematics of experiments or models that you use/introduce, tables showing (hyper)parameters or datasets, etc.

Peer feedback [writing buddy]

Please look at your buddy's draft and provide feedback along the following questions. We will also discuss these questions in the meeting

- Was the intended statement of each topic sentence clear and specific?
- Did the sequence of topic sentences provide a consistent story and build a logical argument? If not, what is missing?
- Was the evidence listed below each topic sentence sufficient to support the statement of the topic sentence? If not, what was missing?
- Do the figure drafts provide the evidence that's listed behind the topic sentences? Are the figures clear and complete? Are they in the right order (as they would be mentioned in the text).
- What are the three main take home messages from the results section?
- Do these take home message fill the gap in knowledge described in the introduction/research proposal?
- Does the outline include subsections for all relevant aspects of the methodology, i.e. data, models, experiments etc.?
- Are there figures illustrating the methodology and do they provide a good overview of the approach and important details?

MILESTONE 3: COMPLETE THESIS DRAFT

The purpose of the third feedback meeting is to make sure you have a complete picture of what your final thesis will look like, so you can concentrate on finalizing a small (!) number of remaining analyses, complete the writing and polish your thesis.

At this point all sections and subsections of the entire thesis should be defined, and all topic sentences should be written. The majority of paragraphs should be drafted or contain at least bullet points of what will be in them.

Peer feedback [writing buddy]

The main objective of this feedback round is checking for completeness. At this point we do not yet care about whether things are written nicely (polishing will be the goal in the remaining time). Answer the following questions:

- Assuming every paragraph will be spelled out, is the thesis complete? If not, what is missing?
- Are all important aspects of the methods, implementation and/or experiments described?
- Results: Do the results present clear evidence for the claims and are the relevant controls done?
- Discussion: Does the discussion address the meaning and implications of the results?